Date
Course

Inequalities and Equations with Decimals

Equations with Decimals

You may use a calculator, and if necessary, round your solutions to the nearest thousandth.

① $2.3x + 5x + 4.6 = -9.41x$

$7.3x + 4.6 = -9.41x$

$-7.3x + 7.3x + 4.6 = -9.41x - 7.3x$

$4.6 = -16.71x$

$\frac{4.6}{-16.71} = \frac{-16.71x}{-16.71}$

$-0.275284 \approx x$

$\boxed{-0.275 \approx x}$

② $-7.6 - 5.8 = \frac{y}{2.6} + 1$

$-13.4 = \frac{y}{2.6} + 1$

$-1 - 13.4 = \frac{y}{2.6} + 1 - 1$

$-14.4 = \frac{y}{2.6}$

$(2.6)(-14.4) = \frac{y}{2.6}(2.6)$

$\boxed{-37.44 = y}$

③ $-2.7z + 3.681 = 4.1z - 10z$

$-2.7z + 3.681 = -5.9z$

$2.7z - 2.7z + 3.681 = -5.9z + 2.7z$

$3.681 = -3.2z$

$\frac{3.681}{-3.2} = \frac{-3.2z}{-3.2}$

$\boxed{-1.15 \approx z}$

④ $\frac{x}{-0.09} - 2.7 = -8.1 + 3$

$\frac{x}{-0.09} - 2.7 = -5.1$

$2.7 + \left(\frac{x}{-0.09}\right) - 2.7 = -5.1 + 2.7$

$\frac{x}{-0.09} = -2.4$

$(-0.09)\frac{x}{-0.09} = -2.4(-0.09)$

$\boxed{x = 0.216}$

⑤ $8y - 0.34y + 3.2 = 7.7y$

$7.66y + 3.2 = 7.7y$

$-7.66y + 7.66y + 3.2 = 7.7y - 7.66y$

$3.2 = 0.04y$

$\frac{3.2}{0.04} = \frac{0.04y}{0.04}$

$\boxed{80 = y}$

⑥ $2.4 - 1 + 0.3 = \frac{z}{3.1} - 4.4$

$1.7 = \frac{z}{3.1} - 4.4$

$4.4 + 1.7 = \frac{z}{3.1} - 4.4 + 4.4$

$6.1 = \frac{z}{3.1}$

$(3.1)\,6.1 = \frac{z}{3.1}(3.1)$

$\boxed{18.91 = z}$

⑦ $-5.1 - 2.2 + 3.888x = 0.56x - 11$

$-7.3 + 3.888x = 0.56x - 11$

$-0.56x - 7.3 + 3.888x = 0.56x - 11 - 0.56x$

$-7.3 + 3.328x = -11$

$7.3 - 7.3 + 3.328x = -11 + 7.3$

$3.328x = -3.7$

$\frac{3.328x}{3.328} = \frac{-3.7}{3.328}$

$x \approx -1.1117885$

$\boxed{x \approx -1.112}$

★ ⑧ $0.006 - \frac{y}{0.04} = 3.8 - 2.1$

$0.006 - \frac{y}{0.04} = 1.7$

$-0.006 + 0.006 - \frac{y}{0.04} = 1.7 - 0.006$

$(-0.04)\left(-\frac{y}{0.04}\right) = 1.694(-0.04)$

$\boxed{y = -0.06776}$

(9) $-\frac{z}{7.8}+1.001=3.8+(-1.1)$

$-\frac{z}{7.8}+1.001=2.7$

$-1.001-\frac{z}{7.8}+1.001=2.7-1.001$

$-\frac{z}{7.8}=1.699$

$(-7.8)\left(-\frac{z}{7.8}\right)=1.699(-7.8)$

$\boxed{z=-13.2522}$

(10) $-3.2-4.5=1.9-\frac{x}{1.08}$

$-7.7=1.9-\frac{x}{1.08}$

$-1.9-7.7=1.9-\frac{x}{1.08}-1.9$

$-9.6=-\frac{x}{1.08}$

$(1.08)(-9.6)=-\frac{x}{1.08}(1.08)$

$-10.368=-x$

$\frac{-10.368}{-1}=\frac{-1x}{-1}$

$\boxed{10.368=x}$

(11) $-5.1(y+2)+3(1.7-4y)=-3-1$

$-5.1y-10.2+5.1-12y=-4$

$-5.1-17.1y=-4$

$-17.1y=1.1$

$y\approx-0.064327$

$\boxed{y\approx-0.064}$

(12) $4x+3.1(2x-3)=1.8(2.5x+1)$

$4x+6.2x-9.3=4.5x+1.8$

$10.2x-9.3=4.5x+1.8$

$5.7x=11.1$

$x\approx1.947368$

$\boxed{x\approx1.947}$

(13) $-1.2(-5.3z+1)=4.3z-8.6$

$6.36z-1.2=4.3z-8.6$

$2.06z=-7.4$

$z\approx-3.5922$

$\boxed{z\approx-3.592}$

(14) $y-2=7.3(-3y-2)$

$y-2=-21.9y-14.6$

$22.9y=-12.6$

$y\approx0.5502$

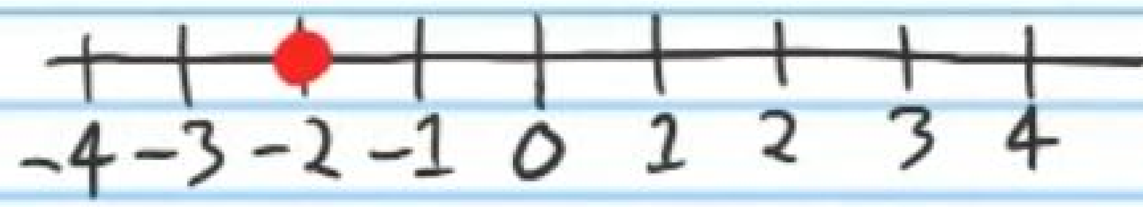
$\boxed{y\approx-0.55}$

Graphing Solutions

Example 1: $-2x-3=7+3x$

Algebraic Solution $\Longrightarrow$ $x=-2$

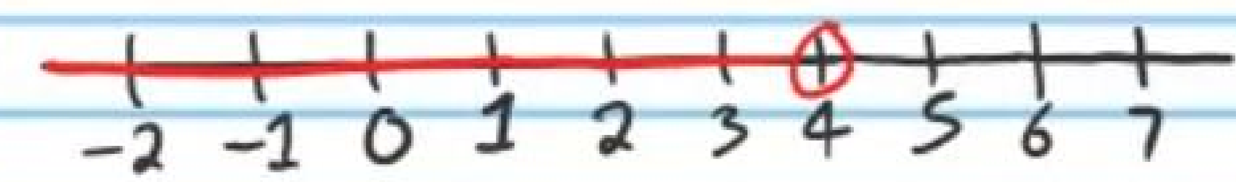
Graphical Solution $\Longrightarrow$



Example 2: $-x+1>2x-11$

Algebraic Solution $\Longrightarrow$ $4>x$

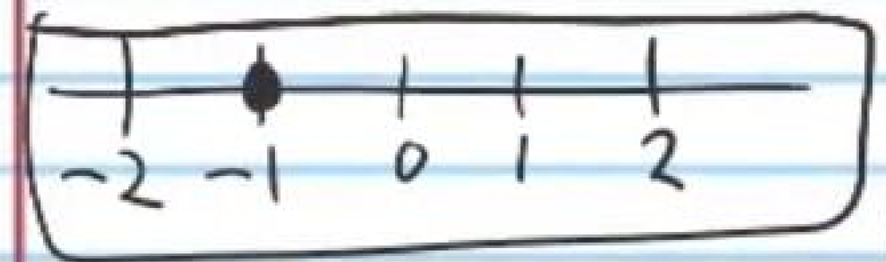
Graphical Solution $\Longrightarrow$



Find the algebraic solutions and graphical solutions of the equations below.

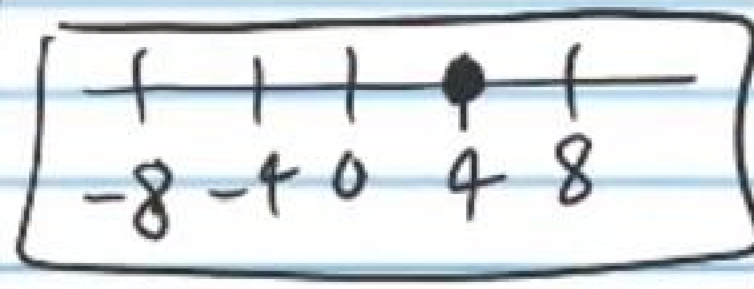
⑮ $2x-1=-3$

$\boxed{x=-1}$



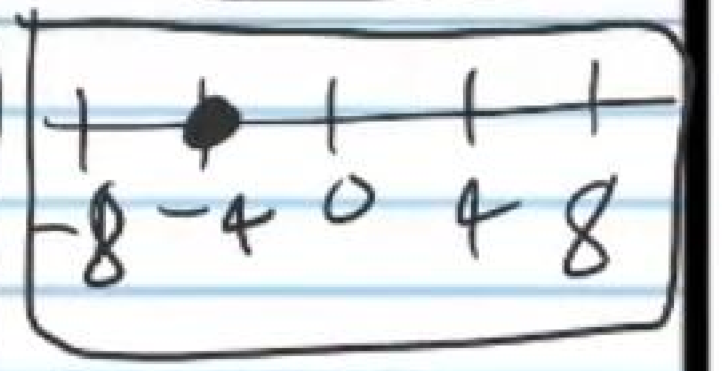
⑯ $-6=\frac{z}{-2}-4$

$\boxed{4=z}$



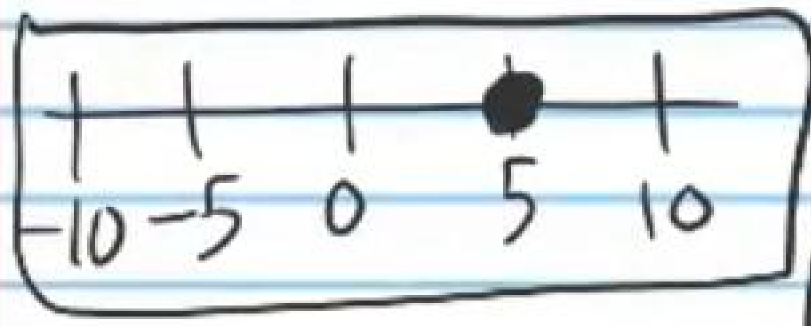
⑰ $3-y=7$

$\boxed{-4=y}$



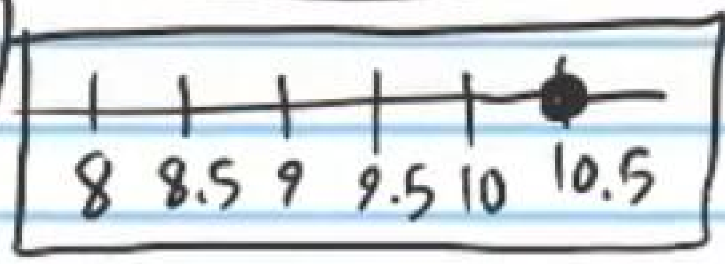
⑱ $2-\frac{x}{5}=1$

$\boxed{x=5}$



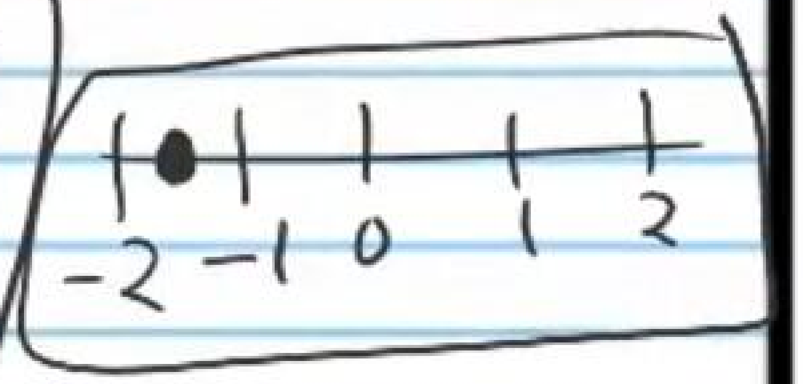
⑲ $2.1=\frac{y}{5}$

$\boxed{10.5=y}$



⑳ $-3x=4.5$

$\boxed{x=-1.5}$

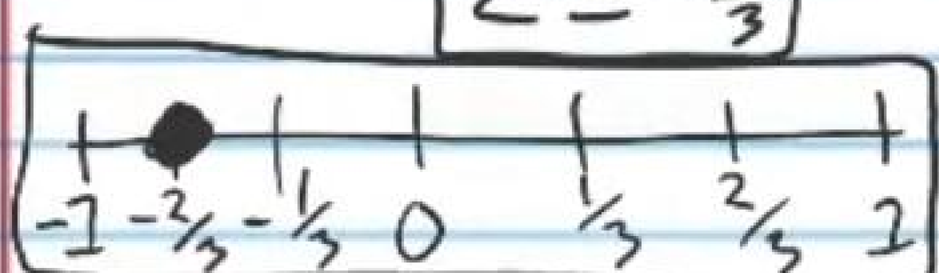


㉑ $5z-2z-5=-7$

$3z-5=-7$

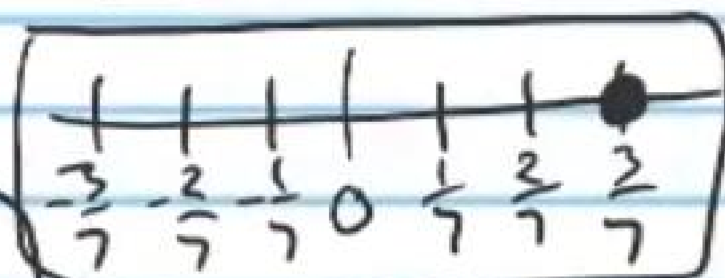
$3z=-2$

$\boxed{z=-\frac{2}{3}}$



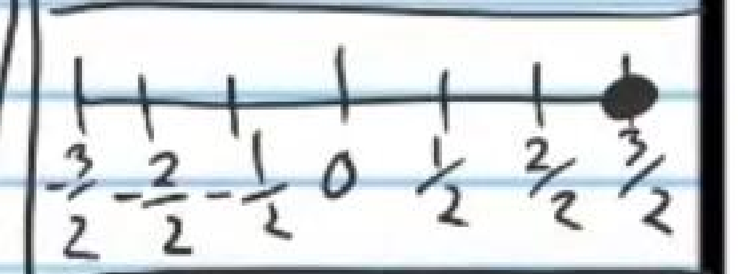
㉒ $-3=-7y$

$\boxed{\frac{3}{7}=y}$



㉓ $2z=3$

$\boxed{z=\frac{3}{2}}$



Inequalities

㉔ $4x+1>8x-3$

$-4x+4x+1>8x-3-4x$

$1>4x-3$

$3+1>4x-3+3$

$4>4x$

$\frac{4}{4}>\frac{4x}{4}$

$1>x$

㉕ $\frac{z}{7}+5<-6+1$

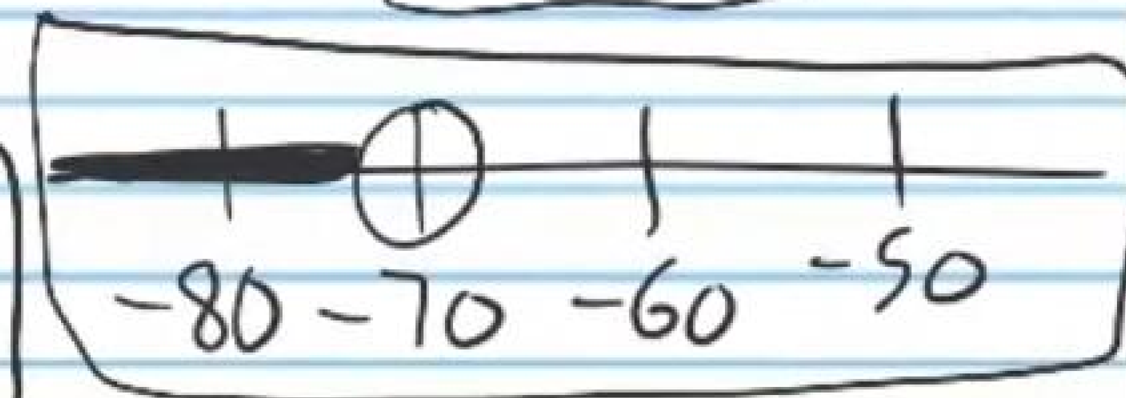
$\frac{z}{7}+5<-5$

$-5+\frac{z}{7}+5<-5-5$

$\frac{z}{7}<-10$

$(7)\frac{z}{7}<-10(7)$

$\boxed{z<-70}$



㉖ $21-3y\leq 1+2y+5y$

$21-3y\leq 1+7y$

$3y+21-3y\leq 1+7y+3y$

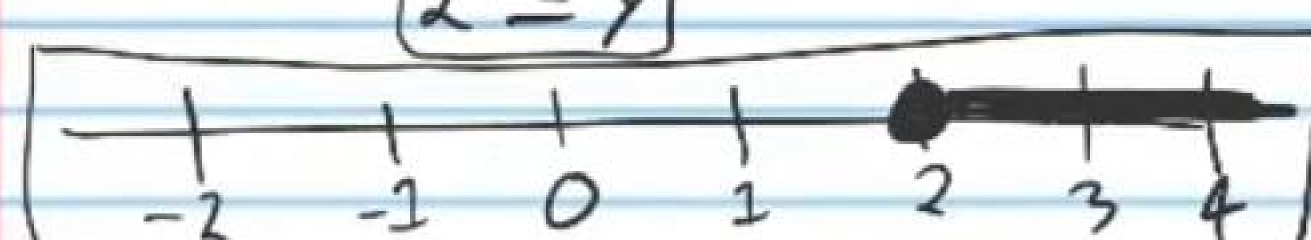
$21\leq 1+10y$

$-1+21\leq 1+10y-1$

$20\leq 10y$

$\frac{20}{10}\leq\frac{10y}{10}$

$\boxed{2\leq y}$



㉗ $1-3-4\geq 5+\frac{x}{2}$

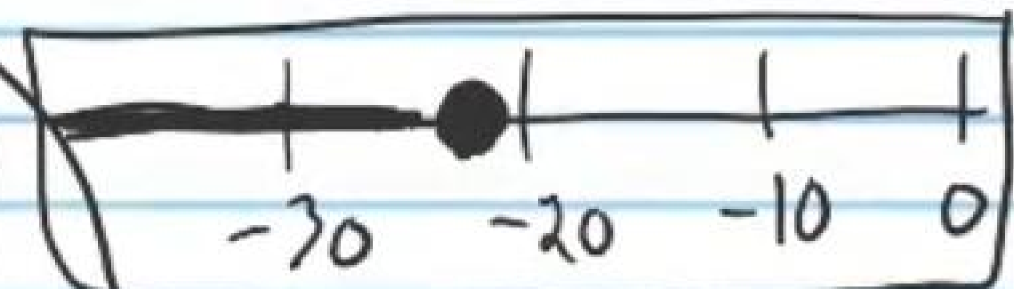
$-6\geq 5+\frac{x}{2}$

$-5-6\geq 5+\frac{x}{2}-5$

$-11\geq\frac{x}{2}$

$(2)(-11)\geq\frac{x}{2}(2)$

$\boxed{-22\geq x}$



(28) $-5y-2y<27+2y$

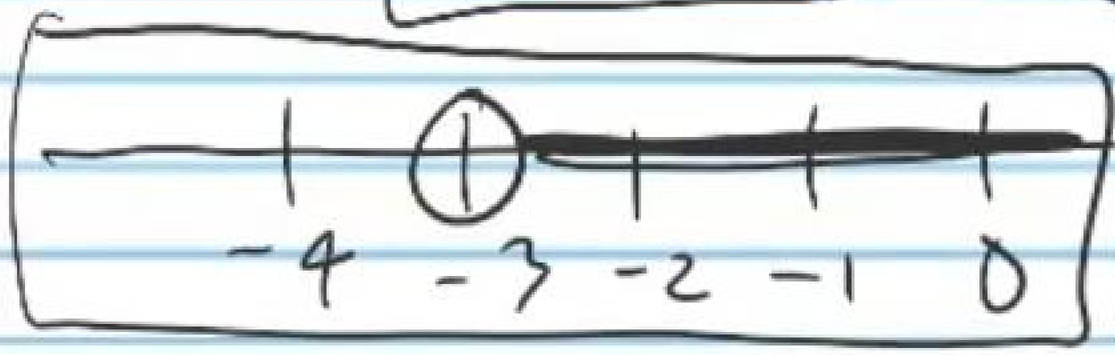
$-7y<27+2y$

$-2y-7y<27+2y-2y$

$-9y<27$

$\dfrac{-9y}{-9}>\dfrac{27}{-9}$

$\boxed{y>-3}$



(29) $1-2\geq\dfrac{z}{-3}+5$

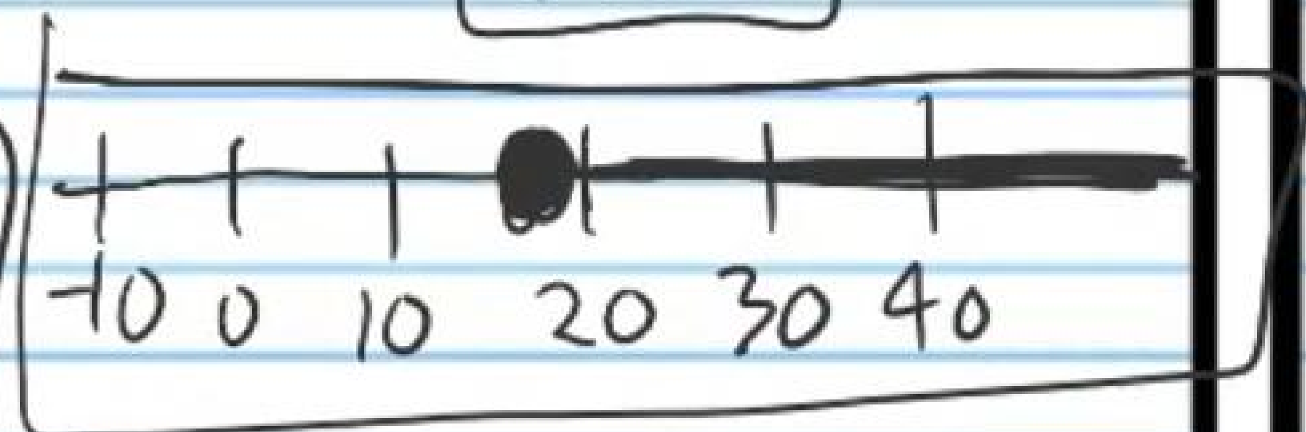
$-1\geq\dfrac{z}{-3}+5$

$-5-1\geq\dfrac{z}{-3}+5-5$

$-6\geq\dfrac{z}{-3}$

$(-3)(-6)\leq\dfrac{z}{-3}(-3)$

$\boxed{18\leq z}$



(30) $-2+x+5<3x-6x$

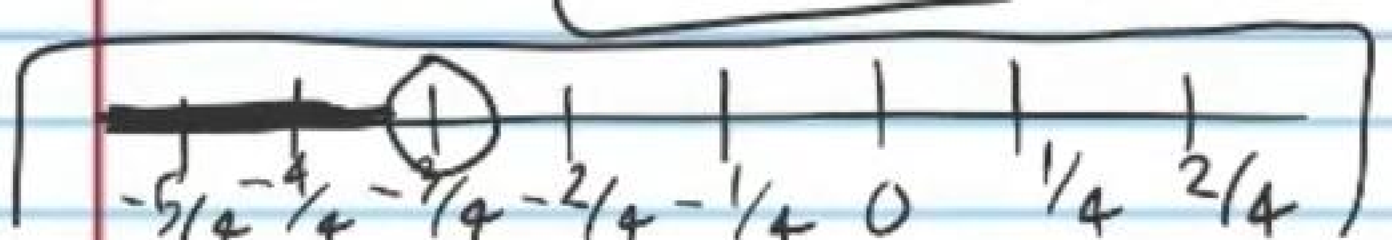
$x+3<-3x$

$-x+x+3<-3x-x$

$3<-4x$

$\dfrac{3}{-4}>\dfrac{-4x}{-4}$

$\boxed{-\dfrac{3}{4}>x}$



(31) $7y+(-10y)\leq-3-y$

$-3y\leq-3-y$

$y-3y\leq-3-y+y$

$-2y\leq-3$

$\dfrac{-2y}{-2}\geq\dfrac{-3}{-2}$

$\boxed{y\geq\dfrac{3}{2}}$

(32) $4+2z+13<17$

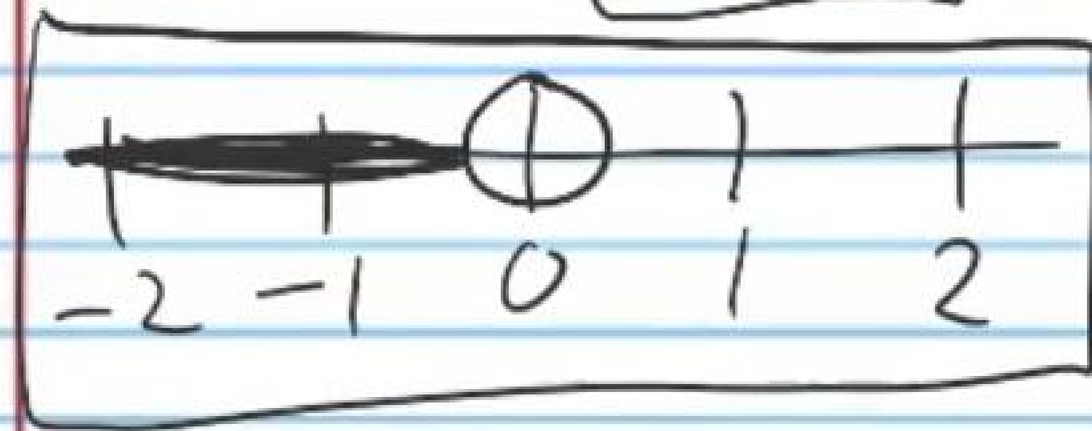
$2z+17<17$

$-17+2z+17<17-17$

$2z<0$

$\dfrac{2z}{2}<\dfrac{0}{2}$

$\boxed{z<0}$



(33) $-6+1\geq3x-5x+9$

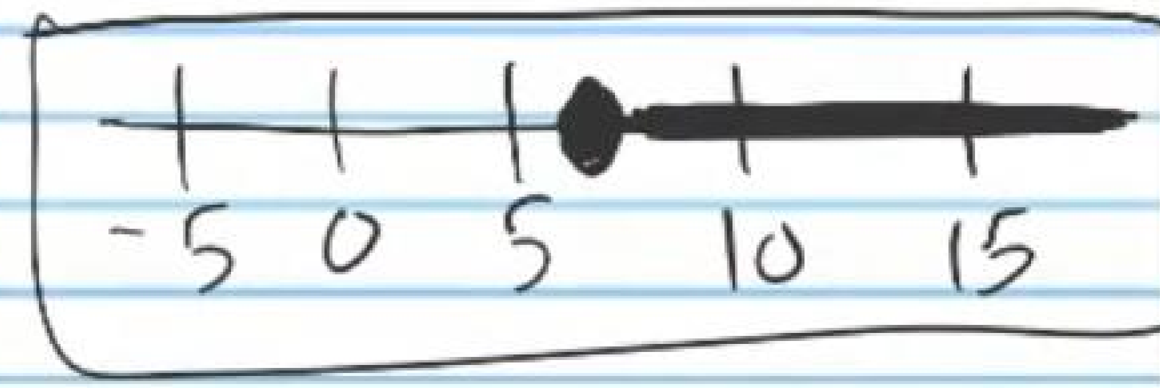
$-5\geq-2x+9$

$-9-5\geq-2x+9-9$

$-14\geq-2x$

$\dfrac{-14}{-2}\leq\dfrac{-2x}{-2}$

$\boxed{7\leq x}$



(34) $-3+\dfrac{y}{-5}>-2+1$

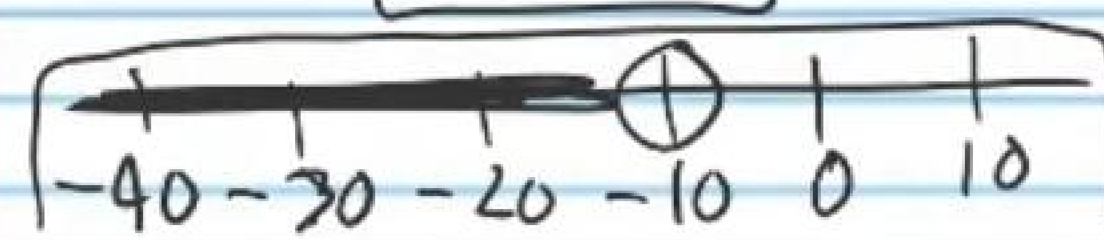
$-3+\dfrac{y}{-5}>-1$

$3-3+\dfrac{y}{-5}>-1+3$

$\dfrac{y}{-5}>2$

$(-5)\dfrac{y}{-5}<2(-5)$

$\boxed{y<-10}$



(35) $-5+5\leq\dfrac{x}{4}+8$

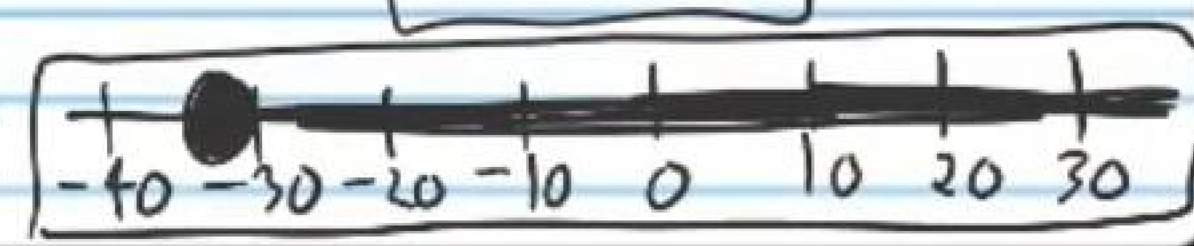
$0\leq\dfrac{x}{4}+8$

$-8+0\leq\dfrac{x}{4}+8-8$

$-8\leq\dfrac{x}{4}$

$(4)(-8)\leq\dfrac{x}{4}(4)$

$\boxed{-32\leq x}$



(36) $-9+4z-9z \ge -2-3$

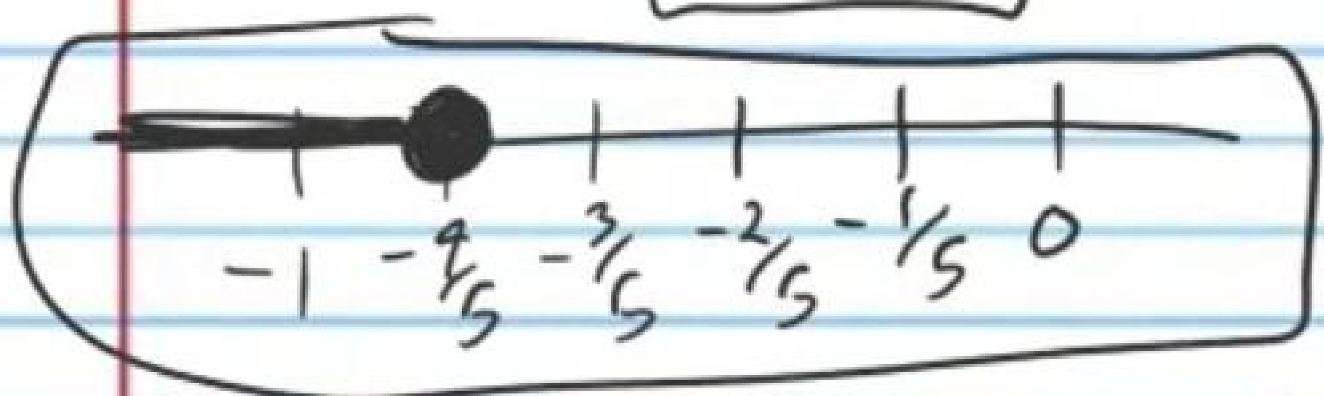
$-9-5z \ge -5$

$9-9-5z \ge -5+9$

$-5z \ge 4$

$\frac{-5z}{-5} \le \frac{4}{-5}$

$\boxed{z \le -\frac{4}{5}}$



(37) $15y-12y > -6+10y$

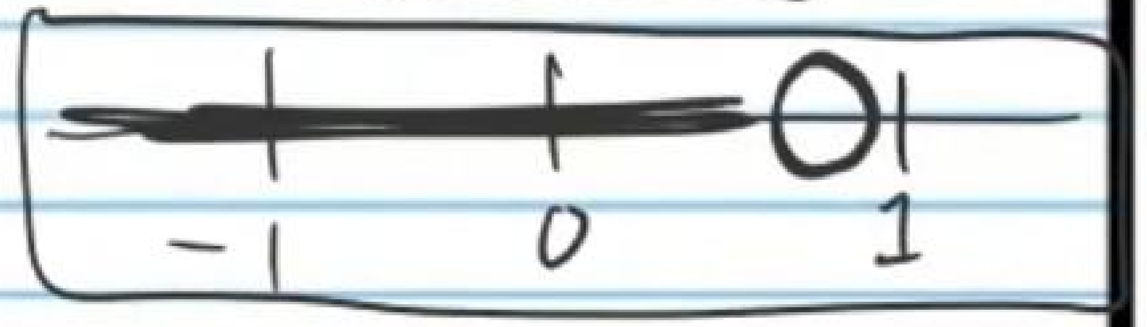
$3y > -6+10y$

$-10y+3y > -6+10y-10y$

$-7y > -6$

$\frac{-7y}{-7} < \frac{-6}{-7}$

$\boxed{y < \frac{6}{7}}$



(38) $4+8x-2x < 25$

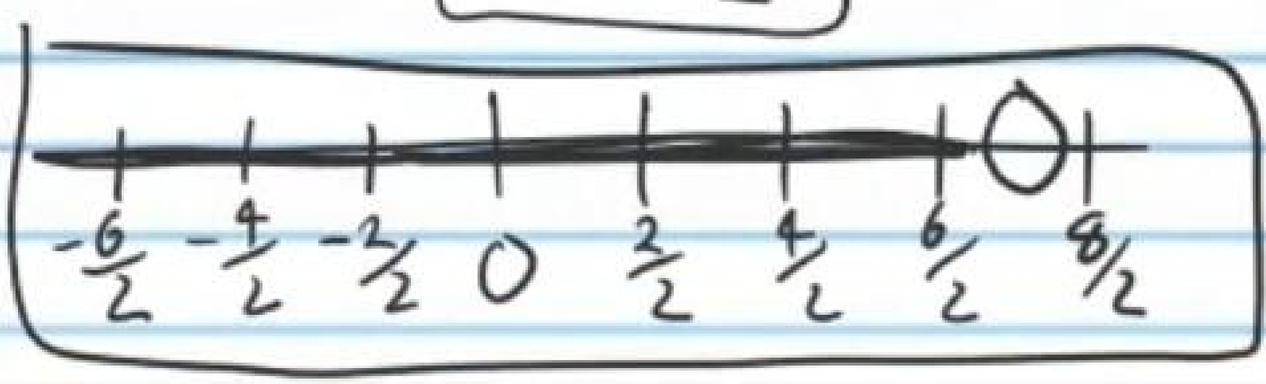
$4+6x < 25$

$-4+4+6x < 25-4$

$6x < 21$

$\frac{6x}{6} < \frac{21}{6}$

$\boxed{x < \frac{7}{2}}$



★(39) $6-\frac{z}{4} > 11-13$

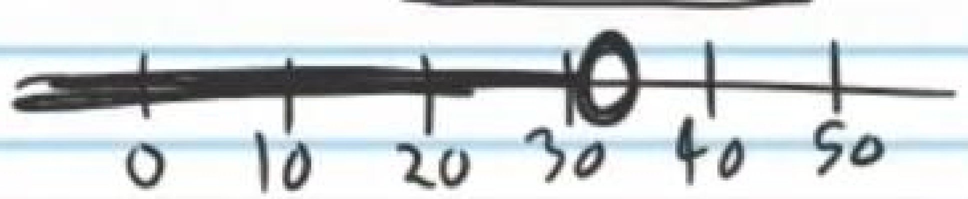
$6-\frac{z}{4} > -2$

$-6+6-\frac{z}{4} > -2-6$

$-\frac{z}{4} > -8$

$(-4)\left(-\frac{z}{4}\right) < -8(-4)$

$\boxed{z < 32}$



(40) $7-8+(-3) \le 2-\frac{x}{7}$

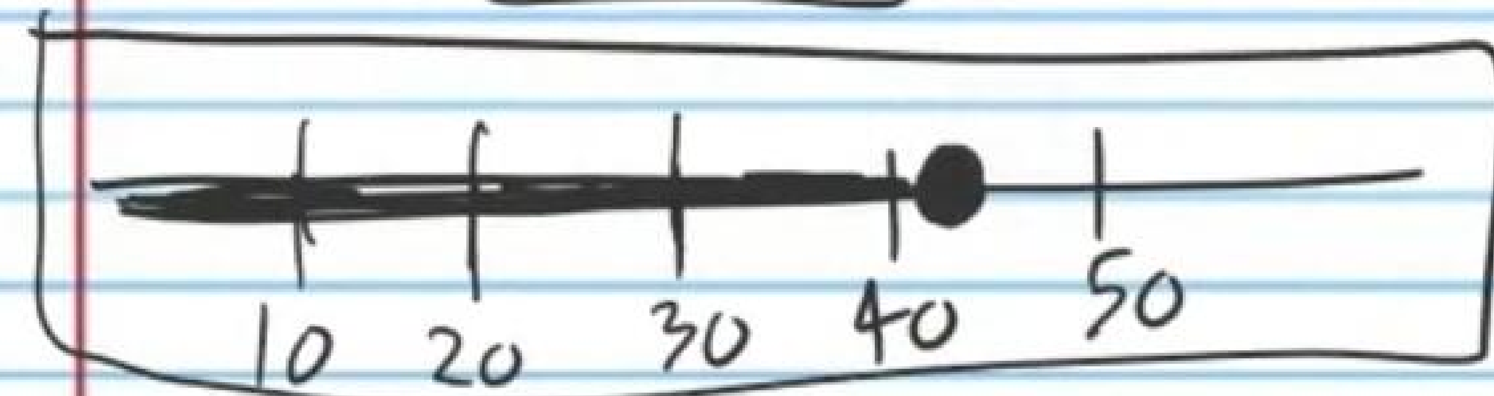
$-4 \le 2-\frac{x}{7}$

$-2-4 \le 2-\frac{x}{7}-2$

$-6 \le -\frac{x}{7}$

$(-7)(-6) \ge -\frac{x}{7}(-7)$

$\boxed{42 \ge x}$



(41) $\frac{y}{2}-3 < 4+(-1)+2$

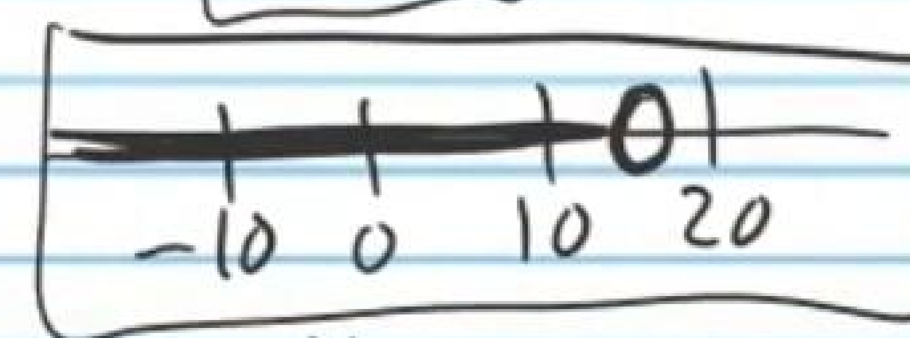
$\frac{y}{2}-3 < 5$

$3+\frac{y}{2}-3 < 5+3$

$\frac{y}{2} < 8$

$(2)\frac{y}{2} < 8(2)$

$\boxed{y < 16}$



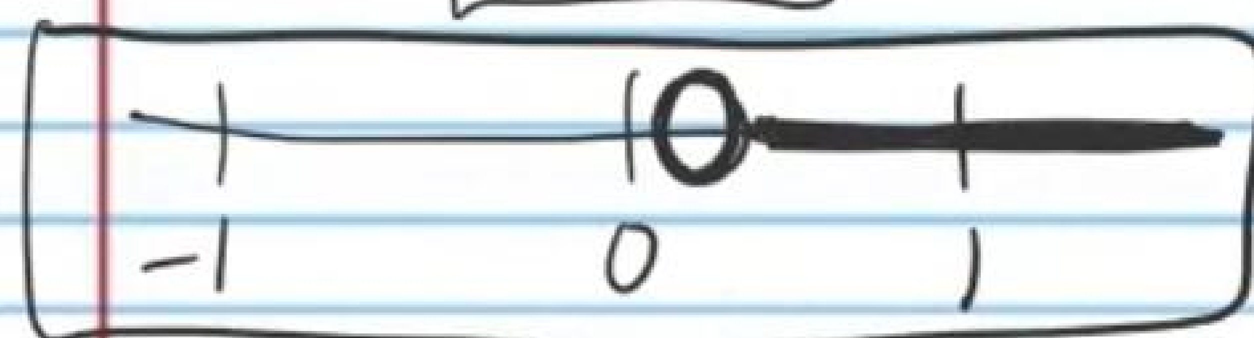
(42) $5z-1 > -3z-4z$

$-5z+5z-1 > -7z-5z$

$-1 > -12z$

$\frac{-1}{-12} < \frac{-12z}{-12}$

$\boxed{\frac{1}{12} < z}$



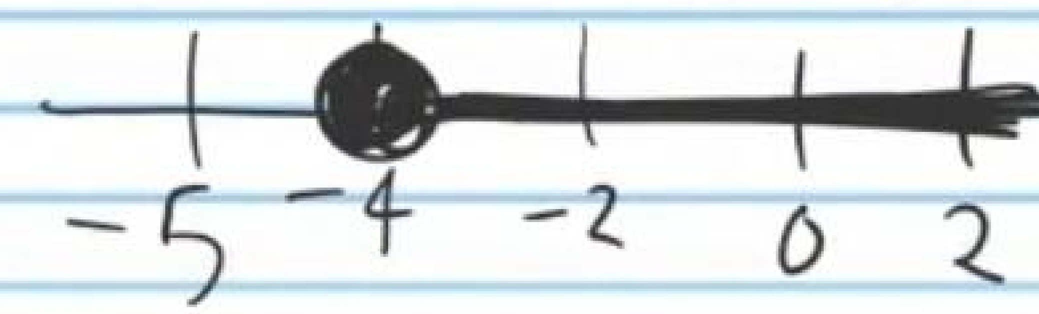
(43) $7+\frac{y}{2} \ge 5$

$-7+7+\frac{y}{2} \ge 5-7$

$\frac{y}{2} \ge -2$

$(2)\frac{y}{2} \ge -2(2)$

$\boxed{y \ge -4}$



44 $-2(x+1)-3 \le 7(1-2x)$

$-2x-2-3 \le 7-14x$

$-2x-5 \le 7-14x$

$14x-2x-5 \le 7-14x+14x$

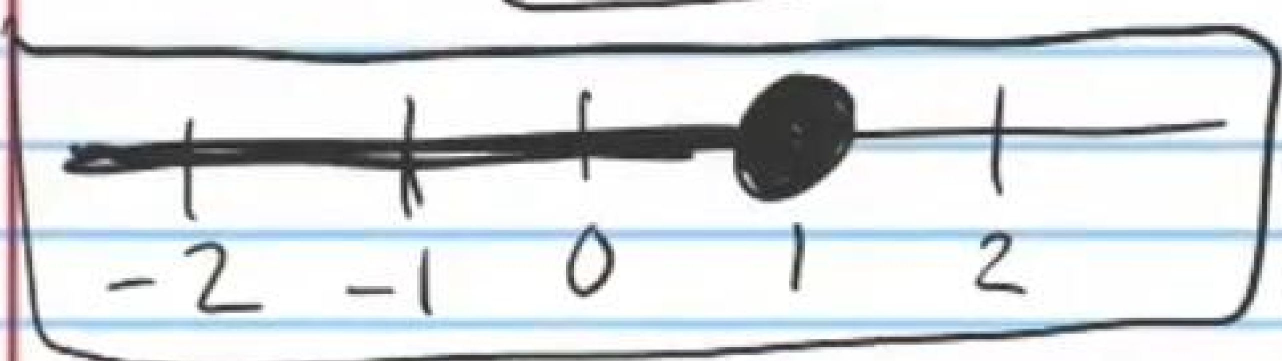
$12x-5 \le 7$

$5+12x-5 \le 7+5$

$12x \le 12$

$\frac{12x}{12} \le \frac{12}{12}$

$\boxed{x \le 1}$



45 $2(-4x+3) > 5-(x+1)$

$-8x+6 > 5-x-1$

$-8x+6 > 4-x$

$8x-8x+6 > 4-x+8x$

$6 > 4+7x$

$-4+6 > 4+7x-4$

$2 > 7x$

$\frac{2}{7} > \frac{7x}{7}$

$\boxed{\frac{2}{7} > x}$